

UNIT 2

PROCESSES IN MACHINE LEARNING

Comprehensive Exam Notes

MCA 521-4 - Machine Learning

Mid-Semester Examination Preparation

Pattern: 5 Questions (3 Compulsory + 2 Optional)

Total Marks: 50

Syllabus Coverage: 2.5 Units

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SECTION A: INFERENCE STATISTICS & HYPOTHESIS TESTING

(Expected Marks: 5-8)

1.1 Population vs Sample

Population: The entire group of individuals or items under study. Sample: A subset of the population selected for analysis. We use sample statistics to infer population parameters.

1.2 Hypothesis Formulation (3 Steps)

IMPORTANT: Very common question! Know all 3 steps clearly.

- Step 1 - State the Problem: Clearly frame the research question and cast it into a testing framework with a defined statistic.
- Step 2 - Calculate Test Statistic: Compute the statistic from sample data. If p-value is very low under the null distribution, reject H_0 .
- Step 3 - Assess Statistical Power: Power = ability to detect a true effect. Depends on significance level (α), sample size (n), and effect size.

NOTE:

Key Rule: If $p\text{-value} < \alpha \rightarrow \text{Reject } H_0$ | Statistical Power = $f(\alpha, n, \text{effect size})$

1.3 Types of Hypotheses

Null Hypothesis (H_0)	Alternative Hypothesis (H_1)
Baseline assumption	What researcher aims to prove
No relationship/difference	Relationship/difference exists
Example: $\text{mean} = \text{mean}_0$	Example: $\text{mean} \neq \text{mean}_0$

1.4 Types of Statistical Models

- Deductive (Statistical/Data Models): Theory-driven, explain underlying causes. Based on General Linear Model (GLM) with normally distributed error terms.
- Inductive (Algorithmic/Data Mining): Accuracy over explanation. Used when processes are too complex to specify - e.g., neural networks, decision trees.
- Extension (Generalized Linear - GLZ): Handles non-normal outcomes: binary (logistic), counts (Poisson). Flexible family of distributions.
- Parametric vs Nonparametric: Parametric assumes known distribution (Gaussian). Nonparametric: no density assumptions; complexity grows with data.

1.5 Types of Error (VERY IMPORTANT)

IMPORTANT: High-yield topic! Know Type I, Type II, Bias, Variance, and Classification errors.

A. Statistical Testing Errors

- Type I Error (α) - FALSE POSITIVE: Rejecting a true H_0 . Concluding a relationship exists when it does not.
- Type II Error (β) - FALSE NEGATIVE: Failing to reject a false H_0 . Missing a real relationship that does exist.

B. Generalization Errors (Bias-Variance)

- Bias: Wrong assumptions $\rightarrow$ Underfitting. Model too simple to learn patterns.
- Variance: Sensitivity to training data $\rightarrow$ Overfitting. Learns noise, not signal.
- Noise: Irreducible error from unobservable variables or imprecise recording.

C. Classification-Specific Errors

- Empirical Error: Proportion of training instances misclassified by the induced hypothesis.
- False Positive (FP): Predicted condition exists when it does not - Type I in classification.
- False Negative (FN): Missed a real condition - critical in clinical diagnosis (cancer screening).

Error Types Summary Table

Error Type	Also Called	Description	Consequence
Type I	False Positive	Reject true H0	False alarm
Type II	False Negative	Fail to reject false H0	Missed detection
High Bias	Underfitting	Model too simple	Poor training & test
High Variance	Overfitting	Model too complex	Good train, poor test

SECTION B: FEATURE TRANSFORMATION & ENGINEERING

(Expected Marks: 5-8)

2.1 Feature Transformation vs Feature Engineering

IMPORTANT: Know the difference! Common comparison question.

Feature Transformation	Feature Engineering
Changes representation	Creates/selects features
Improves format/scale	Improves predictive power
Example: normalization	Example: age from DOB
Focus: Preprocessing	Focus: Model intelligence

2.2 Encoding Techniques

- Nominal Representation: One-hot encoding for unordered categories (colors, countries)
- Ordinal Representation: Integer encoding preserving order (Low=0, Medium=1, High=2)

2.3 Feature Engineering Techniques

- Creating new features from existing ones (e.g., BMI from height and weight)
- Feature selection: Choosing most relevant features
- Feature extraction: PCA, dimensionality reduction
- Binning: Converting continuous to categorical

SECTION C: FEATURE SCALING (NORMALISATION & STANDARDISATION)

(Expected Marks: 5-8)

3.1 Why Feature Scaling?

IMPORTANT: Essential question! Know why scaling is needed.

- Give equal weightage to all features by preventing domination of large-scale features.
- Example: Income (0-100,000) vs Age (0-100) - model may incorrectly favor income just because numbers are larger.
- Essential for distance-based algorithms: KNN, K-means, SVM (RBF kernel) rely on distance calculations.

3.2 Types of Normalisation

1. Min-Max Scaling

$X_{\text{scaled}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$
Result: Range $[0, 1]$

Best for: Algorithms requiring bounded inputs (neural networks), image processing, known data boundaries.

Trade-off: Highly sensitive to outliers - one outlier can drastically expand the range.

2. Mean Normalisation

$X_{\text{scaled}} = (X - \text{mean}) / (X_{\text{max}} - X_{\text{min}})$
Result: Centered around 0, can be negative

Use when: Distribution is unknown or arbitrary, need to center data without rigid bounds.

Trade-off: Even more sensitive to outliers than Min-Max.

3. Max Absolute Scaling

$X_{\text{scaled}} = X / \max(\text{abs}(X))$
Result: Range $[-1, 1]$

Key advantage: Does not shift/center data - preserves sparsity (zero entries remain zero).

Best for: Sparse data (e.g., text data, matrices with mostly zeros).

Trade-off: Sensitive to large outliers.

4. Robust Scaling

$X_{\text{scaled}} = (X - \text{median}) / (Q3 - Q1)$
Uses: Median and IQR (robust statistics)

Best for: Data containing many outliers.

Trade-off: Output range is unbounded, ignores extreme values.

3.3 Standardisation (Z-score Normalisation)

$X_{\text{scaled}} = (X - \text{mean}) / \text{std_dev}$
Result: Mean = 0, Standard Deviation = 1

Best for: Data roughly Gaussian, algorithms assuming normality (e.g., Linear Discriminant Analysis).

3.4 When to Use Which? (Summary)

Method	When to Use	Avoid When
Min-Max	Bounded data, no outliers	Many outliers present
Mean Norm.	Centering needed	Outliers present
Max Abs	Sparse data	Large outliers

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Robust	Many outliers	Need bounded output
Z-score	Gaussian data, normality assumed	Highly skewed data

SECTION D: LINEAR REGRESSION

(Expected Marks: 10-12)

4.1 Regression vs Classification

Aspect	Regression	Classification
Goal	Predict continuous value	Predict category/label
Output	Real numbers	Discrete labels
Algorithms	Linear, Ridge, Lasso	Logistic, DT, SVM
Metrics	MSE, RMSE, MAE, R2	Accuracy, Precision, Recall, F1

4.2 Simple Linear Regression

Finds the best-fitting straight line describing relationship between input (X) and continuous output (Y).

$$Y = B_0 + B_1 \cdot X + \text{error}$$

where B_0 = intercept, B_1 = slope, error = residual

Objective: Minimize the sum of squared residuals (Least Squares Method).

4.3 Least Squares Method

The line that gives the smallest sum of squared residuals (SSR) is the best fit.

- Residual = Actual Y - Predicted Y
- SSR = sum of (residuals)²
- Find slope and intercept that minimize SSR
- One way: Calculate derivative and solve where derivative = 0

4.4 Multiple Linear Regression

Extends simple linear regression by incorporating multiple independent variables.

$$Y = B_0 + B_1 \cdot X_1 + B_2 \cdot X_2 + \dots + B_n \cdot X_n + \text{error}$$

Example: Predicting house price based on size, location, and number of rooms.

4.5 Polynomial Regression

Allows for non-linear relationships by introducing polynomial terms.

$$Y = B_0 + B_1 \cdot X + B_2 \cdot X^2 + B_3 \cdot X^3 + \dots + \text{error}$$

Degree represents the highest power of the input variable.

Polynomial Degree Guide

Degree	Model Shape	Behavior
1	Straight line	Underfitting possible
2	Simple curve	Good for quadratic data
3	Flexible curve	Captures more complexity
High	Very complex curve	Risk of overfitting

IMPORTANT: Small degree -> underfitting | Proper degree -> good generalization | Very high degree -> overfitting

4.6 Evaluation Metrics for Regression

1. Mean Absolute Error (MAE)

$$\text{MAE} = \frac{\sum(|\text{actual} - \text{predicted}|)}{n}$$

Measures average absolute difference. Treats all errors equally. Result in same unit as target variable.

Advantage: Robust to outliers (does not square errors). Simple and easy to interpret.

2. Mean Squared Error (MSE)

$$\text{MSE} = \frac{\sum(\text{actual} - \text{predicted})^2}{n}$$

Average of squared differences. Penalizes larger mistakes more strongly. Sensitive to outliers.

Lower MSE = predictions closer to actual values.

3. Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Square root of MSE. Preferred because error is in same units as original data.

4. R-squared (R²)

$$R^2 = 1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{tot}})$$

Proportion of variance in Y explained by X. Range: 0 to 1 (or negative for poor models).

R² = 1: Perfect fit. R² = 0: Model explains nothing.

4.7 Homoscedasticity vs Heteroscedasticity

IMPORTANT: Important concept! Know the difference and what it means for linear regression.

Homoscedasticity

Same spread of errors. Residuals remain small and evenly spread around regression line for all X values.

$\text{Var}(\text{error}|X) = \text{constant}$

Assumption of linear regression. Required for valid statistical inference.

Heteroscedasticity

Different spread of errors. Residuals become wider as fitted values increase, forming a funnel pattern.

$\text{Var}(\text{error}|X) \neq \text{constant}$

Violates linear regression assumptions. May require transformation or weighted least squares.

SECTION E: MODEL SELECTION (OVERFITTING, UNDERFITTING, BIAS-VARIANCE)

(Expected Marks: 8-10)

5.1 Underfitting

IMPORTANT: Very common question! Know definitions, symptoms, and fixes for both.

- What it is: Model is too simple (e.g., straight line for curved data). Fails to capture underlying trends in BOTH training and testing data.
- Symptoms: High error rates on both training and testing datasets. Model has "high bias".
- How to fix: Increase model complexity (e.g., increase polynomial degree from 1 to 2).

5.2 Overfitting

- What it is: Model is too complex (high-degree polynomial). "Memorizes" training data including noise rather than learning the actual trend.
- Symptoms: Extremely low error on training data, but poor performance and high error on unseen testing data. Model has "high variance".
- How to fix: Reduce model complexity, add more training data, or use regularization (Ridge/Lasso).

5.3 Bias-Variance Trade-off (VERY IMPORTANT)

NOTE:

Central challenge in machine learning! Balance simplicity vs complexity.

- Bias = Blind: Model is too simple to see the pattern. High bias -> underfitting.
- Variance = Volatile: Model changes too much with data. High variance -> overfitting.
- Goal: Low Bias + Low Variance = Best Machine Learning Model.
- Noise: Irreducible error that cannot be eliminated.

Summary Table

Aspect	Underfitting	Overfitting
Model	Too simple	Too complex
Training Error	High	Very low
Testing Error	High	High
Problem	High Bias	High Variance
Fix	Increase complexity	Decrease complexity

SECTION F: DISTANCE MEASURES

(Expected Marks: 5-8)

6.1 Euclidean Distance

Straight-line distance between two points in space. Most common distance metric.

$$d = \sqrt{\sum (x_i - y_i)^2}$$

Best for: Continuous, evenly scaled data. Image recognition, sensor data.

Avoid when: Features vary greatly in scale (use scaling first).

6.2 Manhattan Distance (L1 / Taxicab)

Sum of absolute differences of coordinates. Like a taxi navigating city streets (grid paths).

$$d = \sum (|x_i - y_i|)$$

- Robust to outliers compared to Euclidean
- Preferred for high-dimensional data
- Works well in sparse feature environments

Best for: High-dimensional or grid-based data, city-block routing, clustering.

6.3 Minkowski Distance

Generalized metric that unifies various distance measures.

$$d = (\sum (|x_i - y_i|^p))^{(1/p)}$$

- p=1: Manhattan Distance
- p=2: Euclidean Distance
- p->infinity: Chebyshev Distance

Best for: Need flexible distance tuning with parameter p.

6.4 Cosine Similarity

Measures the angle between two vectors, not magnitude. Captures directional similarity.

$$\text{similarity} = (A \cdot B) / (||A|| * ||B||)$$

- Measures angle, not magnitude
- Good for text or high-dimensional data
- Scale-independent

Best for: Text similarity, embeddings, recommendation systems.

6.5 Distance Metrics Summary

Metric	When to Use	Not Ideal When	Use Case
Euclidean	Continuous, evenly scaled	Features vary in scale	Image recognition
Manhattan	High-dim, grid data	Features correlated	City routing, clustering
Minkowski	Need flexible tuning	Unsure about p value	Generalized KNN
Cosine	Angle matters more	Numerical size matters	Text similarity

SECTION G: K-NEAREST NEIGHBORS (KNN) CLASSIFICATION

(Expected Marks: 8-10)

7.1 What is KNN?

KNN is a lazy learning algorithm that classifies new data points based on the closest existing labeled examples. It relies on distance metrics to quantify similarity among feature values.

- Non-parametric: Makes no assumptions about data distribution
- Lazy learner: No training phase, stores all training data
- Instance-based: Classifies based on similarity to stored instances

7.2 How KNN Works

- Step 1: Choose the number K (number of neighbors)
- Step 2: Calculate distance between query point and all training points
- Step 3: Select K nearest points (smallest distances)
- Step 4: For classification: Majority vote among K neighbors
- Step 5: For regression: Average of K neighbors values

7.3 Choosing K Value

IMPORTANT: Critical parameter! Know how to choose optimal K.

- Small K: Model sensitive to noise -> overfitting (complex model)
- Large K: Smoother boundaries -> may underfit
- Use Cross-Validation to choose best K
- Common practice: $K = \sqrt{n}$ where n = number of samples

7.4 Cross-Validation for K Selection

Technique to check model performance on unseen data while preventing overfitting.

- Split dataset into several parts (folds)
- Train on some parts, test on remaining part
- Repeat with different parts as test set
- Average results for final performance

NOTE:

Stratified Cross-Validation: Maintains class distribution in each fold.

7.5 KNN Performance Metrics

Metric	Meaning
Accuracy	Overall correct predictions
Precision	Correct positive predictions
Recall	Ability to find actual positives
F1 Score	Balance of precision and recall
Confusion Matrix	Detailed prediction comparison

7.6 Limitations of KNN

- Large dataset: Slow prediction (computes distance to all points)
- High dimensions: Distance becomes unreliable (curse of dimensionality)
- Noisy data: Wrong classification possible

- Wrong K value: Overfit or underfit
- No scaling: Incorrect distance calculation
- Stores all data: High memory usage

SECTION H: PERFORMANCE METRICS & CONFUSION MATRIX

(Expected Marks: 8-10)

8.1 Confusion Matrix (VERY IMPORTANT)

IMPORTANT: Foundation of all classification evaluation! Know the 2x2 matrix by heart.

A contingency table that cross-tabulates actual group membership against predicted group membership.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

- True Positives (TP): Positive instances correctly predicted as positive
- True Negatives (TN): Negative instances correctly predicted as negative
- False Positives (FP): Negative instances incorrectly predicted as positive (Type I error)
- False Negatives (FN): Positive instances incorrectly predicted as negative (Type II error)

NOTE:

Diagonal values = correct predictions. Off-diagonal = errors.

8.2 Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Fraction of total samples correctly classified. Most common but misleading in imbalanced datasets.

NOTE:

Example: 95 normal, 5 disease. Model predicts all as normal -> 95% accuracy but fails completely!

8.3 Precision (Positive Predictive Value)

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Of all predicted positives, how many are actually positive?

Prioritize when: Cost of false positive is high. Example: Email spam filters (avoid putting important mail in spam).

8.4 Recall (Sensitivity / True Positive Rate)

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Of all actual positives, how many did we correctly identify?

Prioritize when: Missing a positive case is dangerous. Example: Cancer screening, fraud detection.

8.5 F1 Score

$$\text{F1} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

Harmonic mean of precision and recall. Balances both metrics.

Use when: Need balance between precision and recall. Good for imbalanced datasets.

SECTION I: ROC CURVE & AUC

(Expected Marks: 5-8)

9.1 What is ROC?

IMPORTANT: Know what ROC stands for and what it measures!

ROC = Receiver Operating Characteristic. Plots True Positive Rate (TPR) on Y-axis vs False Positive Rate (FPR) on X-axis for different classification threshold values.

9.2 How to Calculate TPR and FPR

$TPR \text{ (Sensitivity)} = TP / (TP + FN)$

$FPR = FP / (FP + TN)$

9.3 Why ROC? (Why not just accuracy?)

- A model may show high accuracy but still perform poorly, especially with imbalanced data.
- Example: 95 normal, 5 disease. Predict all as normal -> 95% accuracy but completely fails to detect disease.
- ROC evaluates model discrimination capability across ALL thresholds.

9.4 What ROC Tells You

- Compare models: If Model A curve is above Model B, Model A performs better.
- Select best threshold: Lower threshold -> more positives detected, more false positives. Higher threshold -> fewer false positives, may miss true positives.
- Evaluate classifier quality independent of threshold.

9.5 AUC - Area Under the Curve

AUC measures the overall ability of a classifier to distinguish between positive and negative classes.

AUC = Area under the ROC curve

- Larger area -> better classifier
- Smaller area -> poorer classifier

9.6 AUC Interpretation Table

AUC Value	Interpretation
1.0	Perfect classifier
0.9 - 0.99	Excellent
0.8 - 0.89	Good
0.7 - 0.79	Fair
0.5	Random guessing
< 0.5	Worse than random

NOTE:

AUC = 0.5 means the model is no better than flipping a coin.

SECTION J: CROSS-VALIDATION & HYPERPARAMETER TUNING

(Expected Marks: 5)

10.1 Cross-Validation

Technique used to check how well a machine learning model performs on unseen data while preventing overfitting.

- Split dataset into several parts (folds)
- Train model on some parts, test on remaining part
- Repeat process multiple times with different parts as test set
- Average results from each validation step for final performance

10.2 Types of Cross-Validation

- K-Fold Cross-Validation: Split into K equal folds. Use K-1 for training, 1 for testing. Repeat K times.
- Stratified K-Fold: Maintains class distribution in each fold. Important for imbalanced datasets.
- Leave-One-Out (LOO): Each sample is used once as test set. Computationally expensive.

10.3 Hyperparameters vs Model Parameters

Model Parameters	Hyperparameters
Learned from data	Set before training
Example: weights, coefficients	Example: K in KNN, learning rate
Internal to model	External configuration

10.4 Hyperparameter Tuning

- Grid Search: Try all combinations of hyperparameter values. Exhaustive but computationally expensive.
- Random Search: Try random combinations. More efficient than grid search.
- Bayesian Optimization: Uses past results to guide future searches. More intelligent approach.

10.5 Error Analysis

Process of examining errors made by the model to understand failure modes and improve performance.

- Analyze confusion matrix to identify error patterns
- Identify classes that are frequently misclassified
- Determine if errors are systematic or random
- Guide feature engineering and model improvement

APPENDIX A: IMPORTANT FORMULAS

MAE:	$\text{sum}(\text{actual} - \text{predicted}) / n$
MSE:	$\text{sum}((\text{actual} - \text{predicted})^2) / n$
RMSE:	$\text{sqrt}(\text{MSE})$
R-squared:	$1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{tot}})$
Accuracy:	$(\text{TP} + \text{TN}) / \text{Total}$
Precision:	$\text{TP} / (\text{TP} + \text{FP})$
Recall:	$\text{TP} / (\text{TP} + \text{FN})$
F1 Score:	$2 * (\text{P} * \text{R}) / (\text{P} + \text{R})$
TPR/Sensitivity:	$\text{TP} / (\text{TP} + \text{FN})$
FPR:	$\text{FP} / (\text{FP} + \text{TN})$
Euclidean Distance:	$\text{sqrt}(\text{sum}((x_i - y_i)^2))$
Manhattan Distance:	$\text{sum}(x_i - y_i)$
Minkowski Distance:	$(\text{sum}(x_i - y_i ^p))^{(1/p)}$
Cosine Similarity:	$(\text{A} \cdot \text{B}) / (\text{A} * \text{B})$
Min-Max Scaling:	$(X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$
Z-score Standardization:	$(X - \text{mean}) / \text{std_dev}$
Robust Scaling:	$(X - \text{median}) / (\text{Q3} - \text{Q1})$

APPENDIX B: EXAM PREPARATION TIPS

Expected Question Pattern

5 Questions total: 3 Compulsory + 2 Optional (out of given options)

Most Likely Compulsory Questions

Q1: Linear Regression + Evaluation Metrics (MAE, MSE, RMSE, R2)

Expected: 10-12 marks

Q2: Confusion Matrix + Performance Metrics (Accuracy, Precision, Recall, F1)

Expected: 10-12 marks

Q3: Overfitting/Underfitting + Bias-Variance Trade-off + Feature Scaling

Expected: 8-10 marks

Likely Optional Questions

- Option A: KNN Algorithm + Distance Measures + Cross-Validation (8-10 marks)
- Option B: Hypothesis Testing + Types of Error + ROC/AUC (8-10 marks)
- Option C: Feature Engineering + Polynomial Regression + Homoscedasticity (5-8 marks)

Key Diagrams to Practice

- Confusion Matrix (2x2 with TP, TN, FP, FN labels)
- ROC Curve (TPR vs FPR with diagonal reference line)
- Bias-Variance trade-off graph
- Overfitting vs Underfitting illustration
- KNN decision boundary visualization
- Cross-validation folds diagram

Key Definitions to Memorize

- Type I Error (False Positive) vs Type II Error (False Negative)
- Precision vs Recall - when to prioritize each
- Bias vs Variance - definitions and relationship to overfitting/underfitting
- All distance metrics and their formulas
- All feature scaling methods and when to use each
- ROC, AUC, and their interpretation
- KNN algorithm steps and limitations

Python Code Snippets (if asked)

```
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.model_selection import cross_val_score, GridSearchCV

# Feature Scaling
scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

# KNN Classifier
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
predictions = knn.predict(X_test)
```

```
# Confusion Matrix
cm = confusion_matrix(y_test, predictions)

# Cross-Validation
scores = cross_val_score(knn, X, y, cv=5)
```

BEST OF LUCK FOR YOUR MID-SEM EXAM!

You have got this! Prepare well and stay confident.